

^3He gas gap heat switch

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ABSTRACT

Thermal control at 1 K is still demanding for heat switches development.

A gas gap heat switch using ^3He gas as the heat-transfer fluid was tested and characterized. The switch is actuated by a sorption pump, whose triggering temperatures were also characterized. Switching times were recorded for different thermalizations of the sorption pump.

This paper presents the conductance results of such switch. The temperature scanning of the actuator is also presented. The effect of filling pressure is discussed as well as the thermalization of the sorption pump.

About 60 $\mu\text{W/K}$ OFF-state conductance and 100 mW/K ON-state conductance were obtained at 1.7 K. The actuation temperature is slightly adjustable upon the charging pressure of the working gas. Thermalization of the sorption pump at about 8–10 K is enough for producing an OFF state – it can be comfortably linked to a 4 K stage. Temperatures of 15–20 K at the sorption pump are required for reaching the viscous range for maximum ON conduction. Switching time dependence on the thermalization of the sorption pump is discarded.

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1. Introduction

A gas gap heat switch uses an exchange gas between two plates separated by a gap to provide thermal transfer between them, or removal of the gas to reduce the thermal transfer. Actually the ON state is maximized when the gas reaches the viscous regime. At the viscous regime the gas heat transfer is pressure independent and there is no gain upon filling with more gas in the gap. The OFF state is limited by the supporting shell that links both plates. As the gas is pumped off, this supporting shell conductance will dominate the OFF conductance.

The actuation of the gas in the gap is often made using a sorption pump for its ease, lightness and compactness. Such a pump will release the gas to the gap when heated, readsorbing the gas when the pump is cooled down. The thermal management of such a pump may be optimized upon the type and amount of gas to be pumped.

Avoiding liquid ^4He at lower temperatures may be a strong motivation for using ^3He as an exchange gas inside a gas gap heat switch. Besides, ^3He gas has a larger thermal conductivity (in comparison to ^4He) in this range of temperatures.

This paper reports the characterization of a ^3He gas gap heat switch built and tested at the Jet Propulsion Laboratory. Quick switching times were aimed.

ON thermal conductance typically ranges from 0.1 to 2.0 W/K [1] for a ^4He heat switches at 4 K. Several authors report on ^3He gas gap heat switches [2–4]. ON conductance around 0.1 W/K and OFF about 10 $\mu\text{W/K}$ at $T \sim 1.5$ K can be found. One author [2] mentions a turning OFF time of 10 min for a ^3He gas gap heat switch at 0.5 K with about 10 mW/K at ON and 50 $\mu\text{W/K}$ at OFF.

2. Methods

The switch uses two copper blocks with a planar gap of 250 μm width between them (Fig. 1). Total exchange area of 32 cm^2 is available for the gas in the gap. The copper blocks are named the “base” and “top” blocks, the base one being the one that is thermalized to the cold source and the top block the one that is heated up. A stainless steel thin shell supports the copper blocks (49 mm of ϕ 10.9/10.5 mm).

The gas filling of the switch is made at room temperature by a thin ($\phi \sim 1$ mm) copper tube which is pinched off and sealed before cooling.

The switch was tested in a liquid helium cryostat. The base block is thermalized to a 1 K pot. One Cernox thermometer monitors each of the blocks temperature. Top block is also equipped with a heater.

A sorption pump uses about 10 mg of activated charcoal in a separate container. The container is connected to the base block of the switch by means of a stainless steel thin (ϕ 1.0 mm of external diameter, 0.65 mm internal diameter) and long (~ 0.2 m) tube.

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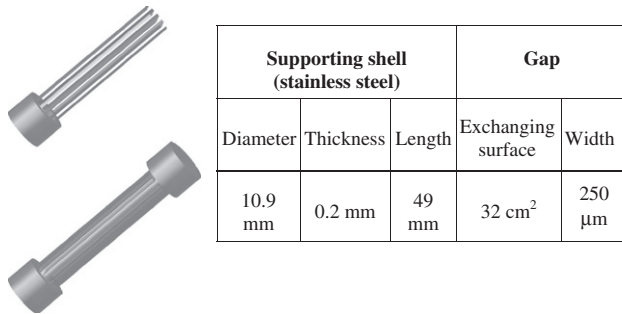


Fig. 1. Schema of the one split block and both blocks assembled for the characterized switch and its main dimensions.

Such a long tube was chosen in order to minimize the power dissipation from the sorption pump to the 1 K pot. Instead of being cooled by the 1 K pot, this tube has a heat intercept to the 4 K stage at $\sim 1/3$ of its length from the pump's side. This way there is a permanent but limited dissipation on the 1 K pot. A diode thermometer and a heater allow the control of the pump.

The measurements are aided by computer recording under a LabView code. Temperatures at the base and top blocks, as well as the temperature of the sorption pump are registered at equilibrium as applied power at the top block and at the sorption pump are stepped.

The equilibrium data was acquired both for a base block temperature of 4.2 K and 1.7 K (with the 1 K pot working). Typical stability criteria of 0.1 K/h were adopted.

Non-equilibrium data was also registered for the ON to OFF transition of the switch. Some heat power is applied to the top block with the switch in the high conductance, then the switch is turned OFF (by letting the sorption pump cool down) and the three temperatures are recorded along time. The purpose of optimization of this ON to OFF time leads us to repeat the measurements for a different heat intercept of the sorption pump. This way a duplicate of the recordings was performed for a heat intercept to the 4 K stage at $1/6$ of the length of the pump's tube. Such a thermalization of the pump should allow it to cool down in half the time, since its time constant halves. Registering the ON to OFF times under different time constants for the cooling down of the pump is expected to help figuring out the dynamics of the gas in the sorbent.

Calculated OFF conductance was performed using the supporting shell geometry and NIST [5] data for the stainless steel conductivity. ON conductance was estimated using a model with parallel elements of copper leave and gas gap in series.

For the supporting shell geometry used, 14 μW are calculated to flow between the cylinder extremities if one is at 1 K and the other at 2 K. The same heat flow would be found through the gas gaps in a molecular regime [6,7] (with an accommodation factor of 1) for a pressure of 10^{-4} Pa. A pressure well below this value is then aimed for minimum OFF conductance.

The mean free path (λ) defines the nature of the heat transfer regime. For $\lambda \gg \text{gap}$ the molecular regime applies, while for $\lambda \ll \text{gap}$ the viscous regime is found. Once the viscous regime is attained, the heat transfer is mostly pressure independent. The minimum pressure of gas that shall ensure the viscous regime was calculated using extrapolated viscosity data for ^3He [8] and calculating the mean free path as $1/100$ of the gap width, using [8,9]:

$$\lambda = 3.6 \frac{\eta}{P} \sqrt{\frac{T}{M}} (SI)$$

For 1 K of temperature a pressure of 15 Pa was estimated, corresponding to 4.5 kPa at room temperature (ideal gas law). The switch was hence tested under two filling pressures of ^3He : 5 kPa

and 50 kPa. The former pressure corresponds approximately to the minimum amount of gas predicted to yield a viscous regime for the ON state of the switch.

3. Results and discussion

Three different types of results are presented: the ON and OFF heat conductance characterization, the effect of sorption pump temperature on the state of the switch, and the turning OFF times.

3.1. Conductance characterization

With a base temperature fixed at 1.7 K and a hot sorption pump, an incremental power is applied to the top block. Once stabilized the temperatures are registered. Applied power as a function of the temperature difference between blocks is plotted (Fig. 2): both filling pressures follow the same linear trend. A linear fit to the experimental results yields a conductance of 96 mW/K. Calculated values match quite well.

OFF state yields large temperature difference and a linear power to temperature difference is no longer seen. This is due to the non-linearity of the integrated thermal conductivity of the stainless steel which increases roughly with T^2 in this range. Experimental OFF conductance of about 0.06 mW/K at 1.7 K can be read from the lower part of the curve plotted in Fig. 3. Corresponding calculated conductance for the supporting shell being 0.03 mW/K. The

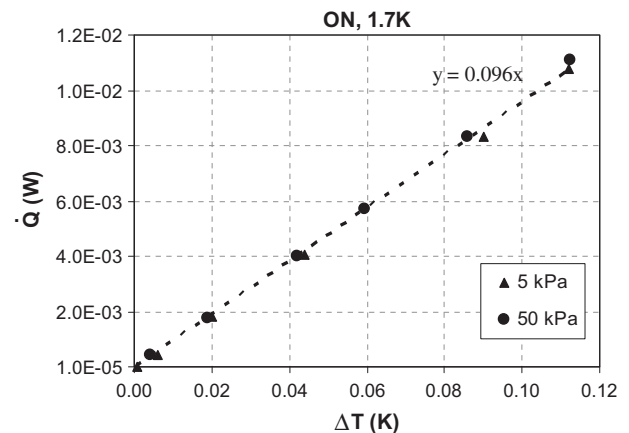


Fig. 2. ON conduction of the switch at 1.7 K.

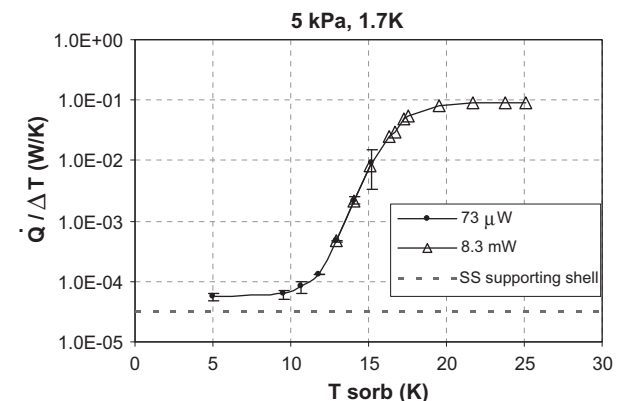


Fig. 3. Sorption pump action on the switch at 1.7 K, while charged with the lowest pressure (5 kPa). Base dashed line represents the calculated conductance for the stainless steel supporting shell.

discrepancy found between OFF conductance values is almost a factor of 2 at 1.7 K. OFF conductance at 1.7 K for 50 kPa does not overlap data for 5 kPa although it is still trending down for lower T_{sorb} . The slight disagreement for OFF conductance at 1.7 K is not fully understood, but it is possible that the sorption pump is not powerful enough to pump the residual gas at 1.7 K. With the base block at 4.2 K the agreement between the experimental OFF conductance and the calculated one for the supporting shell is very good as can be seen in Fig. 5.

3.2. Sorption pump action

Under a low power applied to the top block and a constant base block temperature, the temperature of the sorption pump is stepped. From the large temperature differences obtained with a cold sorption pump, the temperature difference is reduced as the sorption pump temperature is increased. Too large and too narrow temperature differences were avoided by using different values for power as the switch conductance approaches the OFF or the ON side. A constant power of 73 μW for the OFF side (cooler sorption pump side) of Fig. 3 and 8.3 mW for the ON side (warmer sorption pump side) of the same figure was applied. Both powers produce overlapping results in the mid range. Error bars represent the uncertainty in the temperature difference measured under the lower power (solid dots).

In the pressure dependent heat transfer region the effect of the charging pressure can be seen. For higher charging pressure, the pressure in the switch will be higher in the switch for a given pump temperature. The sorption equilibrium is then attained at the same temperature both for a higher sorption amount and higher pressure. Since the heat transfer at the molecular regime is proportional to the pressure, it is expected to yield higher conductance for higher charging pressure.

Different amount of gas in the switch produces a different actuation temperature for the same sorption pump. Fig. 4 compares the actuation temperatures for yielding same ON and OFF conductances under 5 kPa and 50 kPa of room temperature charging pressure. Establishing onset criteria would allow assigning temperatures for the sorption pump to activate the ON and OFF states. For instance, a sorption pump at 10 K leads to a reasonably OFF state of the 5 kPa switch, while the 50 kPa switch still has one order of magnitude higher conductance. The same way, a sorption pump at 15 K leads to a reasonably ON state for the 50 kPa case, while the 5 kPa one is still conducting one order of magnitude less. Depending on the requirements of a particular experiment, those actuation temperatures can be adjusted by choosing the gas-to-charcoal ratio.

Similar results were obtained for a base temperature of 4.2 K, displayed in Fig. 5, with slightly higher conductance both for the ON and OFF state. Indeed there is an increase with temperature of both the conductivities of the viscous gas and metals (copper and stainless steel). As a first approach the gas conductance depends on $T^{1/2}$, which yields a factor of 1.6 from $T = 1.7$ K to $T = 4.2$ K. Such 1.6 factor is very close to what appears to be the difference in ON conductance from Figs. 4 and 5.

Both filling pressures approach the same ON conductance at each base temperature. Such pressure independent conductance demonstrates that the viscous regime was indeed fully established, even for the lower pressure charged switch, as predicted.

3.3. Turning OFF times

Starting with a sorption pump at ~ 25 K (ON viscous state fully established) and some low power applied (0.07 mW) to the top block, the sorption pump's heating is turned off and the recording of temperatures vs. time begins (Fig. 6). The sorption pump itself is

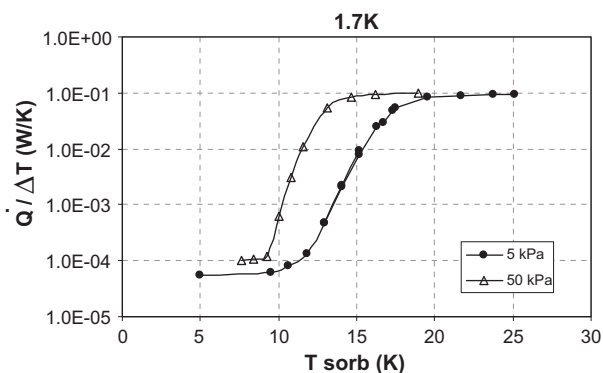


Fig. 4. Sorption pump action on the switch at 1.7 K: its dependence on the charging pressure (5 vs. 50 kPa).

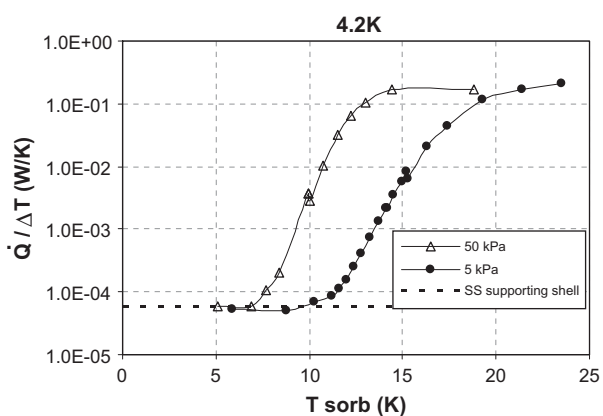


Fig. 5. Sorption pump action on the switch at 4.2 K: its dependence on the charging pressure (5 vs. 50 kPa).

cooling down in 7–20 min, depending on its thermalization. The top block is taking several hours until reaching the equilibrium, independent of the thermalization used. The amount of gas to be pumped (charge pressure of 5 or 50 kPa) does not affect the OFF time. (The slight apparent difference between curves (a) and (b) is just due to a slightly different 1 K pot base temperature achieved. All curves for top block stabilization can be adjusted by an exponential with similar time constants.)

Turning OFF times have shown to be taking several hours. The need for several hours pumping in order to get an open gas gap switch was already mentioned in literature [10]. The slow switching time has been related to the diffusion of the gas in the charcoal [2]. These authors refer that the switching time is much faster with slices of charcoal instead of a single cylinder – they claim this as the key for obtaining the reported 10 min of switching time [2]. Small pieces of charcoal have filled the current sorption pump, and even so the turning OFF times revealed excessively long.

A pumping time constant for the conductance of the tube as been estimated [11] (for the three flow regimes, at 4 K) to be lower than 20 s, discarding its major role to the long OFF times registered.

A thermal time constant (C/K) for the top block of the turning OFF switch has been conservatively estimated to be less than 150 s which is again much lower than the OFF registered times.

Discarding the thermalization influence on the switching OFF speed, one keeps the suspicion on the sorption time problem. The low efficiency of the used adsorbent (low quantity, poor treatment, etc.) may also be limiting the pumping of the gas. The pump itself should be studied.

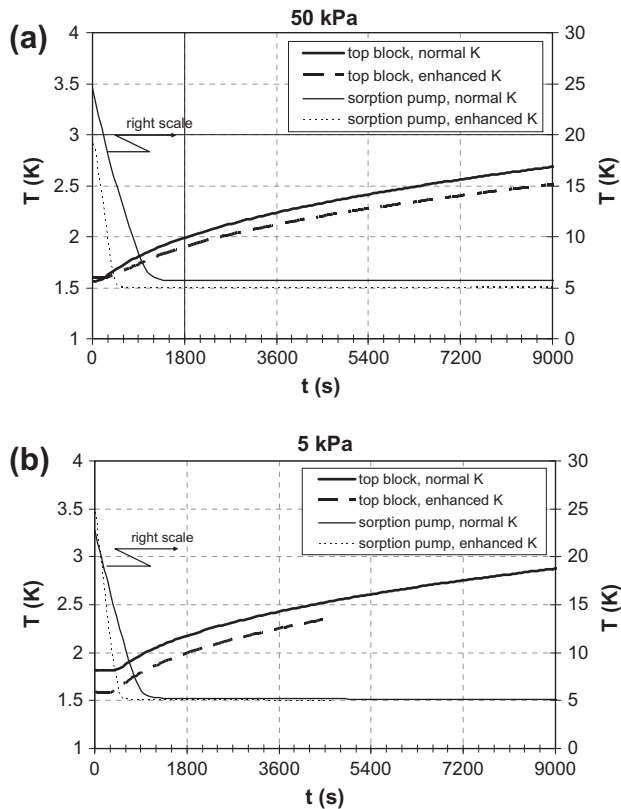


Fig. 6. Switching ON to OFF times: “normal” intercept of the sorption pump to the 4 K stage vs. double conductance for the same intercept. Both charging pressures were recorded, yielding (a) for the highest pressure and (b) for the lowest. Though the sorption pump cools down faster for the improved conductance, there is no faster OFF state found.

4. Conclusion

An ON thermal conductance of 0.1 W/K and an OFF conductance of about 0.06 mW/K were obtained at 1.7 K. A similar OFF conductance was observed at 4.2 K while displaying an ON conductance of 0.2 W/K. Apparently the sorption pump did not pump enough for the lower temperature experiments. ON conductance for both charging pressures is very similar at each temperature, confirming that the viscous regime was indeed achieved even with 5 kPa of gas.

Thermalization of the sorption pump at about 8–10 K is enough for producing an OFF ^3He gas gap heat switch. Such a pump can comfortably be thermalized to a 4 K stage. Heating up to 15–20 K the sorption pump is needed for reaching the viscous ON conduction. The actuation temperature of the charcoal sorption pump is slightly adjustable upon the charging pressure of the gas, as already discussed for other gases [12,13].

Turning ON switching time is rather quick and dependent on the heat power applied to the sorption pump. Turning OFF times have shown to be of several hours. ... Improving the thermal path of the sorption pump to the 4 K stage showed a proportional improved time to cool down the pump, but there is no evidence of its influence on the turning OFF time.

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